<u>Algotunes</u>

1. A short (1 page) write up (pdf) focusing on the technical aspects of your code (or, why we, the TA and I, should be impressed!)
2. A short (1 page) write up that describes a target user of your results and how they would benefit from your work

Algotunes serves as a music recommender, where it takes the input of the user (whether it be an artist or song) and recommends other artists or songs that the user should listen to. It mainly utilizes SpotifyFeatures.csv, which is a large dataset on songs from Spotify. It includes several columns that are a mix of numerical and categorical features, such as genre, popularity, danceability, and key signature.

Our main technical challenge was cleaning the dataset (e.g. dropping obscure artists that had less than 20 songs) and converting the features of the songs/artists into a vector representation. We cleaned the dataset through removing duplicate rows (based on track_id) and keeping the row with the highest popularity and the most complete data. We used OneHotEncoder to convert the categorical features (genre, key, mode, time_signature) so it can be used alongside the numerical features (which was normalized using StandardScaler).

For artist-based recommendations, the system found the mean of the artist's track vectors. We originally tried looping over the artists, which was extremely slow so we used linear algebra instead, building a matrix and computing the artist's sum through matrix multiplication and then the mean through a diagonal inverse matrix. The recommendations were based on cosine similarity since the data here was very high-dimensional. For artist to artist recommendations, it computed the cosine similarity between the artists' vectors. For artist to song recommendations, it took the artist's vector, looked at the songs of a few similar artists, and returned the closest matching songs. For song to song recommendations, it took the song's vector and ranked other tracks by similarity, excluding the same artist for increased variation. We also added a feature so that the recommendations were also affected by the popularity of the track, so that we would avoid obscure picks.

A target user would be someone that wants to explore new songs or artists based on their current musical tastes. For example, if someone really loves listening to "One Dance" by Drake, they can enter the song title and find other similar songs that they would probably enjoy listening to. Or, if they really enjoy a particular artist like Justin Bieber and want to find other songs and artists to listen to as well, they can use Algotunes. This is beneficial because they are able to receive personalized recommendations, increasing the chances that they actually enjoy the new music. Overall, our target user would be someone who wants to expand their musical horizons, whether it be through discovering new songs or artists.